Expectations of function of Joint r.v.

If X and Y have a joint mass/density function $p(x,y)$ then

$$E(g(X,Y)) = \sum_{x,y} g(x,y)p(x,y) \text{ or } \iint_{\mathbb{R}^2} g(x,y)p(x,y)\,dx\,dy.$$

Corollary

$\mathbb{E}(X+Y) = \mathbb{E}(X) + \mathbb{E}(Y)$ *whenever* $\mathbb{E}(X)$ *and* $\mathbb{E}(Y)$ *exist.*

Sample Mean, $\overline{X}$

Let $X_1, \cdots, X_n$ be i.i.d. variables having distribution function F and expected value μ.

$$\mathbb{E}(\overline{X}) = \mathbb{E}\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{1}{n}\mathbb{E}\left(\sum_{i=1}^{n} X_i\right) = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}(X_i) = \mu.$$

Expectation of Products of Independent Random Variables

- If X and Y are independent, then $\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y)$, provided that $\mathbb{E}(X), \mathbb{E}(Y)$ exist.
- If X and Y are independent random variables and g and h are fixed functions, then $\mathbb{E}[g(X)h(Y)] = \mathbb{E}[g(X)] \cdot \mathbb{E}[h(Y)]$, provided that the expectations on the right-hand side exist.

Covariance

If X and Y are jointly distributed random variables with expectations μ_X and μ_Y, respectively, then the covariance of X and Y is

$$Cov(X,Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)] = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y),$$

provided that the expectation exists.

- If X and Y are independent, then $\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y)$ and $Cov(X,Y) = 0$.
- Converse is false, i.e. $Cov(X,Y) = 0 \not\Longrightarrow X$ and Y are independent.

Results from Covariance

- $Cov(a, Y) = 0$.
- $Cov(aX, bY) = ab \cdot Cov(X, Y)$.
- $Cov(aX + bY, Z) = a \cdot Cov(X, Z) + b \cdot Cov(Y, Z)$.
- $Cov(aW + bX, cY + dZ) =$ $ac \cdot Cov(W, Y) + ad \cdot Cov(W, Z) + bc \cdot Cov(X, Y) + bd \cdot Cov(X, Z)$.

Variance between random variables

- $Var(X + Y) = Var(X) + Var(Y) + 2 \cdot Cov(X, Y)$.
- Suppose further that X_i's are independent. Then

$$Var\left(\sum_{i=1}^{n} X_i\right) = \sum_{n=i}^{n} Var(X_i).$$

Correlation Coefficient, ρ

$$\rho = \frac{Cov(X, Y)}{\sqrt{Var(X)Var(Y)}} \qquad -1 \leq \rho \leq 1.$$

Conditional Expectation of Y given $X = x$, $\mathbb{E}(Y|X = x)$

If X and Y are discrete with conditional pmf $p_{Y|X}(y|x)$, then

$$\mathbb{E}(Y|X = x) = \sum_y y p_{Y|X}(y|x).$$

If X and Y are continuous with conditional pdf $f_{Y|X}(y|x)$, then

$$\mathbb{E}(Y|X = x) = \int_y y f_{Y|X}(y|x)\, dy.$$

Conditional Expectation of $g(Y)$ given $X = x$, $\mathbb{E}(g(Y)|X = x)$

Discrete: $\mathbb{E}(g(Y)|X = x) = \sum_y g(y) p_{Y|X}(y|x).$

Continuous: $\mathbb{E}(g(Y)|X = x) = \int_y g(y) f_{Y|X}(y|x)\, dy.$